GeomorphConn

Run Index of Connectivity from DEM, NDVI, and rainfall rasters.

Welcome / About GeomorphConn

What this software does

- Computes hydrologically-weighted Index of Connectivity (IC) maps.
- Produces IC components: IC, Dup, Ddn, W, S, Wmean, Smean.
- Supports D8, D-infinity, and MFD routing options.
- Supports optional aspect-weighted upstream accumulation.
- Handles mismatched input raster grids via automatic alignment.

Core references

- Cavalli et al. (2013), *Geomorphology*, 188, 31-41.
- Crema & Cavalli (2018), *Computers & Geosciences*, 111, 39-45.

Script author

- Manudeo Singh

Method origin credits

- IC methodology: Cavalli et al. (2013)
- SedInConnect reference implementation: Crema & Cavalli (2018)

IC mode

Outlet Target

Flow director

DINF

Slope factor uses dy/dx (percent_rise/100) convention.

Use supplied weight raster (W)

Weight factors

Use aspect weighting

Auto-align rasters

Depression handler

DepressionFinderAndRouter (recommended)

Tip: If flow directions look unusual despite DepressionFinderAndRouter, try providing a DEM that has been pre-filled to remove sinks in GDAL, ArcGIS, or QGIS.

Explicitly fill sinks before routing?

ndvi ×

No (default)

Weight combine

mean

Reference grid

dem

w_min

0.00500

w_max

1.00000

Cache IC outputs for repeated runs

Max cached runs

3

Clear cached outputs before run

DEM GeoTIFF

DTM_201...sin_1m.tif
39.7MB

NDVI GeoTIFF

ndvi_S2.tif
0.9MB

Mask out neighbouring basins (dominant outlet only)

DEM coarsen factor

2× coarser (1/2 resolution, 4× fewer nodes)

Working grid for coarsening: 2835×3667. Available factors: 1×, 2×, 4×, 8×.

Target settings

Target definition method

Flow accumulation threshold

Vector file

Flow direction algorithm: DINF (affects upstream count computation)

Stream generation threshold (upstream cells)

1000

Output settings

Output directory

outputs_gui

Browse...

Output name affix

target_S2NDVI

Output affix mode

suffix

Output layers to save

IC ×

What the output layers mean

- `IC` : final Index of Connectivity, computed as the log-ratio of upstream connectivity potential to downstream impedance.
- `Dup` : upstream component, representing the potential contribution from the upslope contributing area.
- `Ddn` : downstream component, representing the impedance/resistance along the downslope path to the outlet or target.
- `W` : effective weight raster used in the computation after combining selected factors or using a supplied W raster.
- `S` : slope factor used in the IC calculation.
- `Wmean` : mean upstream weight over the contributing area.
- `Smean` : mean upstream slope factor over the contributing area.
- `ACCfinal` : effective contributing cell count used in the D_up averaging (diagnostic).

Save selected outputs to disk

Run IC

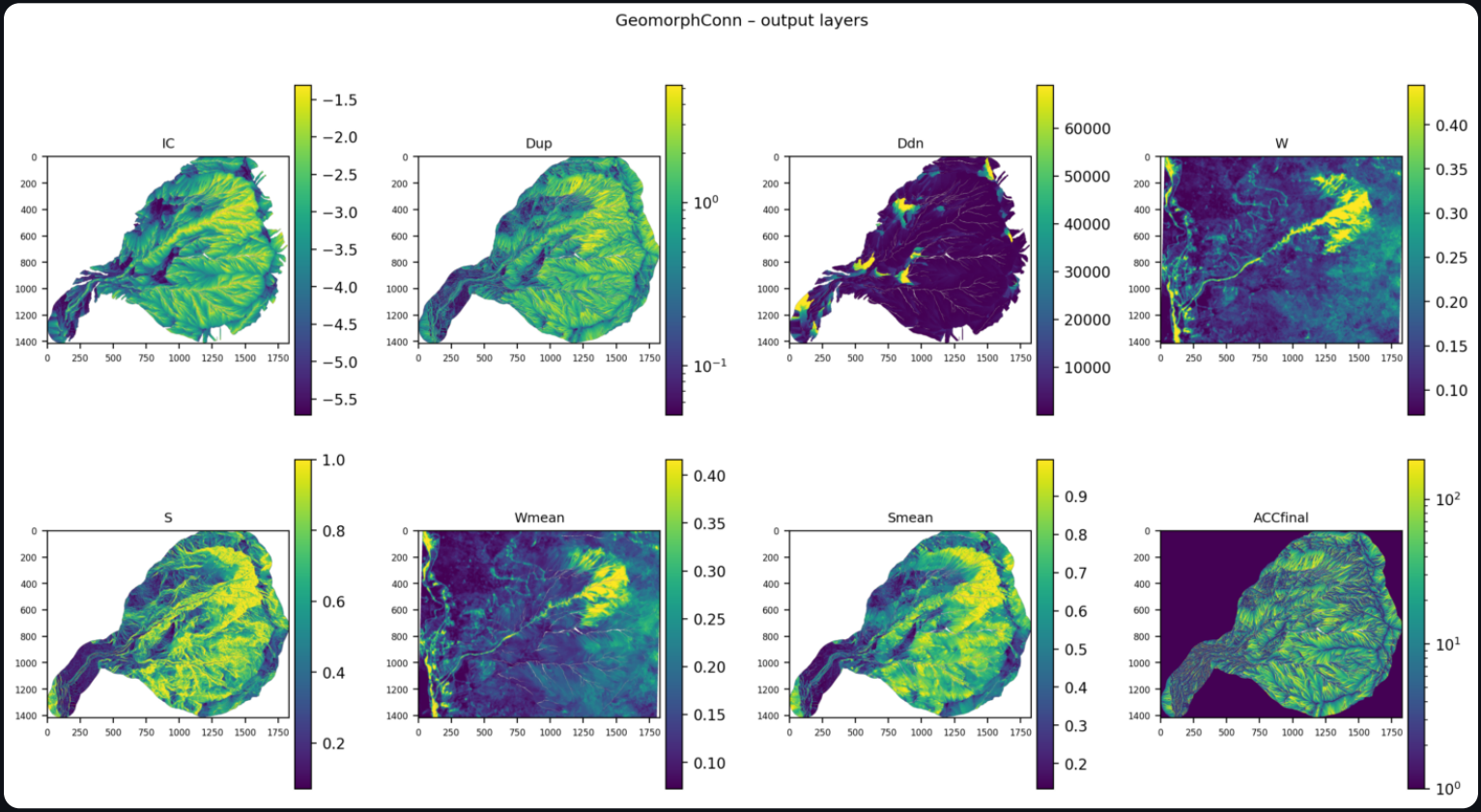
Done.

Large DEM: 10,395,945 nodes (2835×3667). DINF/MFD allocate ~0.6 GiB for internal 8-neighbour arrays. If you hit a MemoryError, increase the Coarsen factor or switch to D8.

IC computation complete.

IC min	IC mean	IC max
-8.019	-3.240	0.593

Output layer maps



Saved run summary: gui_IC_target_S2NDVI.txt

Saved 1 output layer(s) to: outputs_gui